



INDIAN INSTITUTE OF SCIENCE
EDUCATION AND RESEARCH BHOPAL

SIGNALS AND SYSTEMS

Internship Project Booklet

*A comprehensive exploration of signals, systems,
transforms, and frequency-domain analysis*

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MODULE 1

Signals and Systems

Topics Covered in This Module

- ▶ Definition of a Signal and a System
- ▶ Classification of Signals (CT vs DT, Periodic, Even/Odd, Energy & Power)
- ▶ Basic Operations on Signals (Shifting, Scaling, Reversal, Differentiation, Integration)
- ▶ Elementary Signals: Unit Step, Ramp, Impulse, Sinusoids, Exponentials
- ▶ System Properties: Linearity, Time-Invariance, Causality, Stability, Memory, Invertibility
- ▶ Interconnection of Systems (Series, Parallel, Feedback)
- ▶ Solved Examples and GATE-Oriented Problems

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- Haykin, S. & Van Veen, B., *Signals and Systems*, Wiley. **Ch. 1.**
- NPTEL Video Lectures: *Signals and Systems* (IIT Kharagpur / IIT Delhi).
- MIT OCW 6.003: Alan V. Oppenheim online lectures.

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1 Definition of a Signal and a System

What is a Signal?

Definition: Signal

A **signal** is any physical quantity that varies with one or more independent variables (usually time) and carries information about the behaviour or nature of some phenomenon.

Mathematically, a signal is represented as a *function* of one or more independent variables. For example:

- $x(t)$ — a **continuous-time (CT)** signal; $t \in \mathbb{R}$
- $x[n]$ — a **discrete-time (DT)** signal; $n \in \mathbb{Z}$

Real-world examples of signals:

Signal	Physical Quantity	Type
Speech	Acoustic pressure vs. time	CT
ECG	Heart's electrical activity vs. time	CT
Image	Pixel intensity vs. (x, y) position	2-D CT
Stock price	Price vs. trading day	DT
Sampled audio	Amplitude at discrete samples	DT
Temperature sensor	Temperature vs. time	CT
DNA sequence	Base pair vs. position index	DT (spatial)

Note: Multi-Dimensional Signals

Signals need not be functions of time alone. An image $I(x, y)$ is a **2-D signal** (two spatial variables). A video $V(x, y, t)$ is a **3-D signal**. The course primarily deals with 1-D time-domain signals, but the same principles generalise.

What is a System?

Definition: System

A **system** is any process or device that accepts one or more input signals and produces one or more output signals in response. It performs some *transformation* or *processing* on the input.

$$\underbrace{x(t)}_{\text{Input}} \xrightarrow{\mathcal{T}\{\cdot\}} \underbrace{y(t)}_{\text{Output}}$$

Examples of systems:

- An *RC circuit* — input voltage $v_s(t)$, output capacitor voltage $v_c(t)$.
- An *amplifier* — input audio, output amplified audio.
- An *automobile* — input engine force $f(t)$, output velocity $v(t)$.
- A *digital filter* — input sampled sequence $x[n]$, output filtered sequence $y[n]$.
- A *communication channel* — input transmitted signal, output received (possibly distorted) signal.

Note: Notation Convention

- Continuous-time signals use **round brackets**: $x(t)$, $y(t)$
- Discrete-time signals use **square brackets**: $x[n]$, $y[n]$
- This distinction is **strictly maintained** throughout the subject.

Interconnection of Systems

Real-world systems are rarely isolated. They are composed of simpler subsystems connected in various ways. Understanding interconnections is fundamental.

Series (Cascade) Connection

The output of one system becomes the input of the next.

$$x(t) \longrightarrow \boxed{\mathcal{T}_1} \longrightarrow w(t) \longrightarrow \boxed{\mathcal{T}_2} \longrightarrow y(t)$$

$$y(t) = \mathcal{T}_2\{\mathcal{T}_1\{x(t)\}\}$$

Example: A pre-amplifier followed by a power amplifier.

Parallel Connection

Both systems receive the same input; their outputs are summed.

$$y(t) = \mathcal{T}_1\{x(t)\} + \mathcal{T}_2\{x(t)\}$$

Example: A bank of filters processing the same signal simultaneously.

Feedback Connection

Part of the output is fed back and combined with the input.

$$e(t) = x(t) - y(t), \quad y(t) = \mathcal{T}\{e(t)\}$$

Example: An operational amplifier with negative feedback; cruise control in a car.

Why Feedback Matters

Feedback is used to improve stability, reduce sensitivity to component variations, and shape the frequency response of a system. It is foundational in control theory.

2 Classification of Signals

2.1 Continuous-Time vs. Discrete-Time Signals

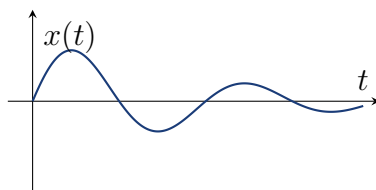
Continuous-Time (CT) Signal

A signal $x(t)$ defined for **every** value of t in a given interval (often $-\infty < t < \infty$) is called a **continuous-time signal**. The independent variable is continuous.

Discrete-Time (DT) Signal

A signal $x[n]$ defined **only at integer values** of n ($n \in \mathbb{Z}$) is called a **discrete-time signal**. It is often obtained by *sampling* a CT signal at uniform intervals T_s : $x[n] = x(nT_s)$.

Continuous-Time Signal



Discrete-Time Signal

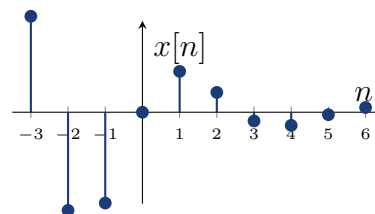


Figure 1. Continuous-Time signal $x(t)$ (left) and Discrete-Time signal $x[n]$ (right).

Feature	CT Signal	DT Signal
Notation	$x(t), t \in \mathbb{R}$	$x[n], n \in \mathbb{Z}$
Defined at	Every real instant of time	Only integer instants
Source	Natural/analog phenomena	Sampled or inherently digital data
Representation	Smooth curve	Stem (lollipop) plot
Processing	Differential equations	Difference equations

Common Mistake: “Discrete” vs. “Digital”

A **discrete-time** signal has a continuous (infinite-precision) amplitude but is defined only at integer time instants. A **digital** signal is both discrete-time *and* discrete-amplitude (quantised). In this course, “discrete-time” does NOT imply quantisation unless stated.

2.2 Periodic and Aperiodic Signals

Periodic Signal (CT)

A CT signal $x(t)$ is **periodic** with period $T > 0$ if

$$x(t) = x(t + T), \quad \forall t \in \mathbb{R}.$$

The *smallest* such T is the **fundamental period** T_0 ; $f_0 = 1/T_0$ is the **fundamental frequency** (Hz), and $\omega_0 = 2\pi/T_0$ is the **fundamental angular frequency** (rad/s).

Periodic Signal (DT)

A DT signal $x[n]$ is **periodic** with period N (positive integer) if

$$x[n] = x[n + N], \quad \forall n \in \mathbb{Z}.$$

The smallest such integer N is the **fundamental period** N_0 .

Periodicity of $\cos(\omega_0 t)$ in CT vs. DT [GATE/ESE]

- **CT:** $x(t) = \cos(\omega_0 t)$ is *always* periodic, $T_0 = 2\pi/\omega_0$, for any $\omega_0 \neq 0$.
- **DT:** $x[n] = \cos(\omega_0 n)$ is periodic *only if* $\omega_0/(2\pi) = m/N$ is a *rational* number. If irrational, the DT signal is **aperiodic**!

DT fundamental period formula:

$$N_0 = \frac{2\pi m}{\omega_0}$$

where m is the *smallest positive integer* that makes N_0 a positive integer.

Solved Example 1: DT Periodicity [GATE/ESE]

Q. Is $x[n] = \cos\left(\frac{3\pi}{7}n\right)$ periodic? If so, find N_0 .

Solution: Here $\omega_0 = 3\pi/7$.

$$\frac{\omega_0}{2\pi} = \frac{3\pi/7}{2\pi} = \frac{3}{14} \quad (\text{rational})$$

So the signal is **periodic**. Take $m = 3$, $N = 14$ (lowest integers):

$$N_0 = \frac{2\pi \cdot 3}{3\pi/7} = \frac{6\pi \cdot 7}{3\pi} = 14$$

$$\boxed{N_0 = 14}$$

Q. Is $x[n] = \cos(\sqrt{2}n)$ periodic?

$\omega_0/(2\pi) = \sqrt{2}/(2\pi)$, which is **irrational** $\Rightarrow$ **not periodic**.

Solved Example 2: Period of Sum of Sinusoids [GATE/ESE]

Q. Find the fundamental period of $x(t) = \cos(2t) + \sin(3t)$.

Solution: $T_1 = 2\pi/2 = \pi$, $T_2 = 2\pi/3$.

$$T_0 = \text{LCM}(T_1, T_2) = \text{LCM}\left(\pi, \frac{2\pi}{3}\right) = 2\pi$$

Method: Write $T_1/T_2 = (\pi)/(2\pi/3) = 3/2$ (rational) $\Rightarrow$ signal is periodic. $T_0 = 2\pi$.
Verify: $\cos(2(t + 2\pi)) = \cos(2t + 4\pi) = \cos(2t) \checkmark$; $\sin(3(t + 2\pi)) = \sin(3t + 6\pi) = \sin(3t) \checkmark$.

2.3 Even and Odd Signals**Even and Odd Signals**

A signal $x(t)$ is called **even** if $x(-t) = x(t)$ for all t .

A signal $x(t)$ is called **odd** if $x(-t) = -x(t)$ for all t .

Important observations:

- $\cos(\omega_0 t)$ is **even**; $\sin(\omega_0 t)$ is **odd**.
- $e^{j\omega_0 t}$ is **neither** (complex-valued).
- The only signal that is both even and odd is $x(t) = 0$.
- For an odd signal: $\int_{-\infty}^{\infty} x_o(t) dt = 0$ (if the integral exists).

Even–Odd Decomposition

Every signal can be uniquely decomposed as:

$$x(t) = \underbrace{\frac{x(t) + x(-t)}{2}}_{\mathcal{E}v\{x(t)\} = x_e(t) \text{ (even part)}} + \underbrace{\frac{x(t) - x(-t)}{2}}_{\mathcal{O}d\{x(t)\} = x_o(t) \text{ (odd part)}}$$

Solved Example 3: Even–Odd Decomposition [GATE/ESE]

Q. Find the even and odd parts of $x(t) = e^{-t}u(t)$.

Solution: Note $x(-t) = e^t u(-t)$.

$$x_e(t) = \frac{e^{-t}u(t) + e^t u(-t)}{2} = \frac{1}{2}e^{-|t|}$$

$$x_o(t) = \frac{e^{-t}u(t) - e^t u(-t)}{2} = \frac{1}{2}e^{-|t|} \text{sgn}(t)$$

where $\text{sgn}(t) = +1$ for $t > 0$ and -1 for $t < 0$.

Verification: $x_e(t) + x_o(t) = \frac{1}{2}e^{-|t|}(1 + \text{sgn}(t))$. For $t > 0$: $= \frac{1}{2}e^{-t}(2) = e^{-t} = x(t) \checkmark$.
For $t < 0$: $= 0 = x(t) \checkmark$.

2.4 Energy and Power Signals

Total Energy and Average Power — CT

$$E_{\infty} = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$P_{\infty} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

Total Energy and Average Power — DT

$$E_{\infty} = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Signal Class	E_{∞}	P_{∞}	Example
Energy Signal	$0 < E_{\infty} < \infty$	$P_{\infty} = 0$	$e^{-at}u(t)$, $a > 0$
Power Signal	$E_{\infty} = \infty$	$0 < P_{\infty} < \infty$	$A \cos(\omega_0 t)$, $u(t)$
Neither	$E_{\infty} = \infty$	$P_{\infty} = \infty$	e^{at} , $a > 0$; ramp $r(t)$

Key Rules for Energy/Power Classification [GATE/ESE]

- **Finite-duration, finite-amplitude** signals $\Rightarrow$ always **Energy** signals.
- **Periodic** signals $\Rightarrow$ always **Power** signals.
- A signal **cannot** be both energy and power simultaneously.
- A signal **can** be neither (e.g. e^t for $t \in \mathbb{R}$).
- $u(t)$ is a **power signal**: $P_{\infty} = 1/2$.
- $\delta(t)$ has $E = 1$ (under a suitable limiting definition) — treated as energy signal.

Power of Standard Signals [GATE/ESE]

$$x(t) = A \cos(\omega_0 t + \phi): \quad P_{\infty} = \frac{A^2}{2}$$

$$x(t) = Ae^{j\omega_0 t}: \quad P_{\infty} = |A|^2$$

$$x(t) = A \cos(\omega_1 t) + B \cos(\omega_2 t), \omega_1 \neq \omega_2: \quad P_{\infty} = \frac{A^2}{2} + \frac{B^2}{2}$$

$$x(t) = A \text{ (DC)}: \quad P_{\infty} = A^2$$

Solved Example 4: Energy of a Decaying Exponential [GATE/ESE]

Q. Find the energy of $x(t) = e^{-2t}u(t)$.

Solution:

$$E = \int_0^{\infty} (e^{-2t})^2 dt = \int_0^{\infty} e^{-4t} dt = \left[\frac{e^{-4t}}{-4} \right]_0^{\infty} = 0 - \left(-\frac{1}{4} \right) = \frac{1}{4} \text{ J}$$

Since $0 < E < \infty$ and $P = 0$, this is an **energy signal**.

$$E = 0.25 \text{ J}$$

3 Basic Operations on Signals

Overview

Signal operations act either on the **independent variable** (time axis transformations) or on the **dependent variable** (amplitude transformations). Mastering these is essential for convolution, Fourier, and Laplace analysis.

3.1 Time-Domain Operations

1. Time Shifting

Time Shift

$$y(t) = x(t - t_0)$$

- $t_0 > 0$: signal is **delayed** (shifted right along t -axis).
- $t_0 < 0$: signal is **advanced** (shifted left along t -axis).

2. Time Reversal (Reflection)

Time Reversal

$$y(t) = x(-t)$$

The signal is **flipped** about the vertical axis $t = 0$.

3. Time Scaling

Time Scaling

$$y(t) = x(at), \quad a > 0$$

- $|a| > 1$: signal is **compressed** (sped up in time).
- $0 < |a| < 1$: signal is **expanded** (slowed down in time).
- $a < 0$: includes both scaling *and* reversal.

Common Mistake: Direction of Scaling

Students often confuse compression and expansion. Remember:

- $x(2t)$: period halves $\Rightarrow$ **compressed** (shorter, faster).
- $x(t/2)$: period doubles $\Rightarrow$ **expanded** (wider, slower).

Intuition: multiplying t by a large number makes the signal “rush through” its cycle faster.

4. Amplitude Scaling and Addition

$$y(t) = Ax(t), \quad z(t) = x_1(t) + x_2(t)$$

Amplitude scaling stretches or compresses the signal vertically. Addition is the basis of **superposition** in linear systems.

5. Differentiation and Integration (CT)

These are amplitude transformations of practical importance:

Differentiation and Integration

$$y(t) = \frac{d}{dt}x(t), \quad y(t) = \int_{-\infty}^t x(\tau) d\tau$$

Key relationships: $\frac{d}{dt}u(t) = \delta(t)$; $\frac{d}{dt}r(t) = u(t)$; $\int_{-\infty}^t \delta(\tau) d\tau = u(t)$.

These are repeatedly exploited in circuit analysis (capacitor current $i = C dv/dt$, inductor voltage $v = L di/dt$).

6. Combined Transformation $x(at + b)$ **Order-of-Operations for $x(at + b)$ [GATE/ESE]**

To obtain $y(t) = x(at + b)$ from $x(t)$, use **either** approach:

Method A (shift first, then scale):

1. Shift $x(t)$ by b/a : gives $x(t + b/a)$.
2. Scale by a : replace t with at , gives $x(at + b)$.

Method B (scale first, then shift):

1. Scale $x(t)$ by a : gives $x(at)$.
2. Shift by $-b/a$: replace t with $t + b/a$, gives $x(a(t + b/a)) = x(at + b)$.

Both methods are equivalent. **Never** just substitute directly without reasoning about what the axis is doing.

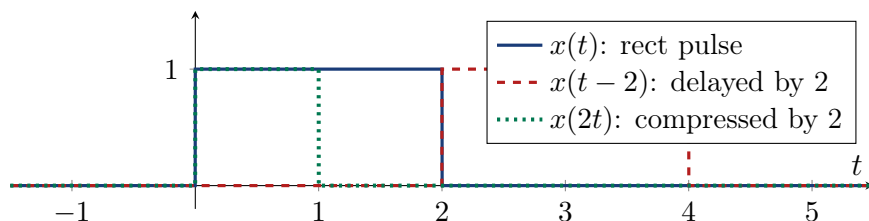


Figure 2. Time Shifting and Time Scaling on a rectangular pulse $x(t) = \text{rect}(t/2 - 1/2)$.

Solved Example 5: Combined Transformation [GATE/ESE]

Q. Given $x(t)$: a pulse from $t = 1$ to $t = 3$ of amplitude 1. Sketch $y(t) = x(2t - 4)$.

Solution: Write $2t - 4 = 2(t - 2)$. So scale first ($x(2t)$: pulse from $t = 0.5$ to $t = 1.5$), then shift right by 2 ($x(2(t - 2))$: pulse from $t = 2.5$ to $t = 3.5$).

Or: shift first by $4/2 = 2$ left ($x(t + 2)$? – no). Careful: $x(2t - 4)$, so $b = -4$, $a = 2$, shift by $-b/a = 2$ to the right. Original pulse $x(t)$: $1 \leq t \leq 3$. New limits: $1 \leq 2t - 4 \leq 3 \Rightarrow 5 \leq 2t \leq 7 \Rightarrow 2.5 \leq t \leq 3.5$. $y(t) = 1$ for $2.5 \leq t \leq 3.5$, zero otherwise.

4 Elementary Signals

4.1 Unit Step Function $u(t)$ and $u[n]$

Unit Step Function

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \quad u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Uses: Switching on a signal, expressing one-sided signals, defining piecewise functions:

$$x(t) \cdot u(t - t_0) \text{ turns on } x(t) \text{ at } t = t_0.$$

Important: $u(t)$ is a **power signal** with $P_\infty = 1/2$.

Expressing Piecewise Signals using $u(t)$

Any piecewise-constant signal can be written as a sum of shifted step functions.

Example: A pulse of amplitude 2 from $t = 1$ to $t = 3$:

$$x(t) = 2[u(t - 1) - u(t - 3)]$$

This technique is extremely useful for computing Laplace transforms of piecewise signals.

4.2 Unit Ramp Function $r(t)$

Unit Ramp Signal

$$r(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases} = t \cdot u(t)$$

Integral relationship:

$$r(t) = \int_{-\infty}^t u(\tau) d\tau \quad \Longleftrightarrow \quad \frac{d}{dt} r(t) = u(t)$$

4.3 Unit Impulse (Dirac Delta) Function

Unit Impulse $\delta(t)$

The Dirac delta is defined via the **sifting property**:

$$\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0)$$

Informally: $\delta(t) = 0$ for $t \neq 0$ and $\int_{-\infty}^{\infty} \delta(t) dt = 1$.

Intuition: $\delta(t)$ as a Limiting Pulse

The Dirac delta can be understood as the limit of a rectangular pulse of width ϵ and height $1/\epsilon$ as $\epsilon \rightarrow 0$:

$$\delta(t) = \lim_{\epsilon \rightarrow 0} \begin{cases} 1/\epsilon & 0 \leq t \leq \epsilon \\ 0 & \text{otherwise} \end{cases}$$

The area is always 1, but the pulse becomes infinitely tall and infinitely narrow. It is *not* an ordinary function — it is a **distribution**.

Key properties of $\delta(t)$:

Property	Expression / Explanation
Sifting	$\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0)$
Scaling	$\delta(at) = \frac{1}{ a } \delta(t)$ (area preserved)
Relation to step	$u(t) = \int_{-\infty}^t \delta(\tau) d\tau, \quad \frac{d}{dt}u(t) = \delta(t)$
Even symmetry	$\delta(-t) = \delta(t)$
Generalized scaling	$\delta(f(t)) = \sum_k \frac{\delta(t - t_k)}{ f'(t_k) }$, where $f(t_k) = 0$
Product	$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$

Common Mistake to Avoid

$\delta(t)$ is **not** an ordinary function. Never evaluate $\delta(0)$ as a number. Always use the **sifting property** in calculations.

Also, $\delta(2t) \neq \delta(t)$. Using the scaling property: $\delta(2t) = \frac{1}{2} \delta(t)$.

Solved Example 6: Impulse Properties [GATE/ESE]

(a) Evaluate $\int_{-\infty}^{\infty} (t^2 + 3) \delta(t - 2) dt$.

By sifting: $= (2^2 + 3) = 7$. 7

(b) Simplify $\delta(3t - 6)$.

$$3t - 6 = 3(t - 2), \text{ so } \delta(3t - 6) = \delta(3(t - 2)) = \frac{1}{3}\delta(t - 2).$$

(c) Evaluate $\int_{-\infty}^{\infty} e^{-t} \cos(t) \delta(t + 1) dt$.

By sifting at $t_0 = -1$: $= e^{-(-1)} \cos(-1) = e \cos(1) \approx 1.468$.

4.4 Unit Impulse in Discrete Time

Unit Sample (Kronecker Delta) $\delta[n]$

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

Sifting property: $\sum_{n=-\infty}^{\infty} x[n] \delta[n - n_0] = x[n_0]$

Useful identities:

$$u[n] - u[n - 1] = \delta[n], \quad u[n] = \sum_{k=0}^{\infty} \delta[n - k]$$

CT vs. DT Impulse: Key Difference

- DT unit sample $\delta[n]$ is a **perfectly ordinary function** with a concrete value of 1 at $n = 0$. No mathematical complications.
- CT Dirac delta $\delta(t)$ is a **distribution** with no finite value at $t = 0$. Must be handled via integration.

4.5 Complex Exponential Signals

General Complex Exponential

Continuous-time:

$$x(t) = Ce^{at}, \quad C, a \in \mathbb{C}$$

Discrete-time:

$$x[n] = C\alpha^n, \quad C, \alpha \in \mathbb{C}$$

Let $C = |C|e^{j\theta}$ and $a = \sigma + j\omega_0$ (in CT), so $Ce^{at} = |C|e^{\sigma t} \cos(\omega_0 t + \theta) + j|C|e^{\sigma t} \sin(\omega_0 t + \theta)$.

Special Cases and Physical Interpretation

Case	CT: a value	Waveform	DT: α
Pure real, positive	$a = \sigma > 0$	Growing exponential	$ \alpha > 1$
Pure real, negative	$a = \sigma < 0$	Decaying exponential	$0 < \alpha < 1$
Pure imaginary	$a = j\omega_0$	Sinusoid (constant amp.)	$ \alpha = 1$
Complex	$a = \sigma + j\omega_0, \sigma < 0$	Damped sinusoid	$ \alpha < 1$
Complex	$a = \sigma + j\omega_0, \sigma > 0$	Growing sinusoid	$ \alpha > 1$

Physical Importance of Complex Exponentials

The complex exponential $e^{j\omega_0 t}$ is the **eigenfunction** of every LTI system. This means that if the input to an LTI system is $e^{j\omega_0 t}$, the output is simply $H(j\omega_0) e^{j\omega_0 t}$ — the same complex exponential scaled by a (possibly complex) constant $H(j\omega_0)$. This property is the foundation of **Fourier analysis** and frequency-domain methods.

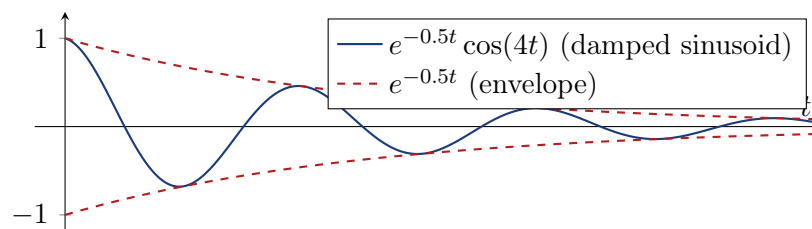


Figure 3. Damped sinusoid $e^{-0.5t} \cos(4t)$ with exponential envelope. This arises naturally in RLC circuits and mechanical oscillators.

Euler's Formula

Euler's Formula (Most Important Identity in Signals & Systems)

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Why it matters: Euler's formula lets us represent all sinusoidal signals as sums of complex exponentials. Since complex exponentials are eigenfunctions of LTI systems, *all* Fourier and Laplace analysis ultimately rests on this identity.

Solved Example 7: Representing a Sinusoid [GATE/ESE]

Q. Express $x(t) = 3 \cos(2t + \pi/4)$ using complex exponentials.

$$x(t) = 3 \operatorname{Re}\{e^{j(2t+\pi/4)}\} = \frac{3}{2} (e^{j(2t+\pi/4)} + e^{-j(2t+\pi/4)}) = \frac{3}{2} e^{j\pi/4} e^{j2t} + \frac{3}{2} e^{-j\pi/4} e^{-j2t}$$

The two-sided spectrum has components at $\omega = +2$ and $\omega = -2$, each with amplitude

$3/2$ and phases $\pm\pi/4$.

5 Properties of Systems

Why System Properties Matter

Classifying a system by its properties tells us which mathematical tools apply. For example: the **convolution integral** applies only to **LTI** systems; the **BIBO stability** test involves the impulse response; causal systems admit real-time implementation. GATE/ESE questions on system properties are among the most frequently tested topics in Signals & Systems.

5.1 Linearity

Linear System

A system $\mathcal{T}\{\cdot\}$ is **linear** if and only if it satisfies **superposition**:

$$\mathcal{T}\{ax_1(t) + bx_2(t)\} = a\mathcal{T}\{x_1(t)\} + b\mathcal{T}\{x_2(t)\}$$

for all inputs x_1, x_2 and all constants $a, b \in \mathbb{C}$. This combines:

1. **Homogeneity:** $\mathcal{T}\{ax(t)\} = ay(t)$
2. **Additivity:** $\mathcal{T}\{x_1(t) + x_2(t)\} = y_1(t) + y_2(t)$

Quick Test for Linearity [GATE/ESE]

1. **Zero input $\Rightarrow$ zero output?** If $x(t) = 0$ gives $y(t) \neq 0$ (non-zero initial conditions or DC bias), the system is **non-linear**.
2. **Non-linear terms present?** Any $y^2, |y|, y \cdot x, \sin(y), e^y, \sqrt{y}$ make the system non-linear.
3. **Affine system:** $y = mx + c$ with $c \neq 0$ is **not linear** (it's affine, not linear).
4. **All terms linear:** A system described by a linear ODE with *zero* initial conditions is linear.

Solved Example 8: Testing Linearity [GATE/ESE]

(a) $y(t) = t \cdot x(t)$

$\mathcal{T}\{ax_1 + bx_2\} = t(ax_1 + bx_2) = a(tx_1) + b(tx_2) = ay_1 + by_2$. **Linear.** ✓

(b) $y(t) = x^2(t)$

$\mathcal{T}\{ax_1\} = (ax_1)^2 = a^2x_1^2 \neq ax_1^2$. Homogeneity fails. **Non-linear.** ✗

(c) $y(t) = 2x(t) + 3$

With zero input: $y = 3 \neq 0$. **Non-linear** (affine). ✗

(d) $\frac{dy}{dt} + 2y(t) = x(t)$ (with zero initial conditions)

This is a linear ODE with constant coefficients and zero IC. **Linear.** ✓

5.2 Time-Invariance (Shift-Invariance)

Time-Invariant System

A system is **time-invariant** if a delay in the input causes an equal delay in the output, with no other change:

$$x(t) \rightarrow y(t) \implies x(t - t_0) \rightarrow y(t - t_0) \quad \forall t_0 \in \mathbb{R}$$

Quick Test for Time-Invariance [GATE/ESE]

1. t appears explicitly (outside of $x(\cdot)$)? E.g. $t \cdot x(t)$, $e^{-t}x(t) \Rightarrow$ **time-varying**.
2. **Time-scaled input?** $y(t) = x(2t)$ (compressor) $\Rightarrow$ **time-varying**. Even though no explicit t multiplies x .
3. **Constant-coefficient linear ODE** (with x appearing linearly)? $\Rightarrow$ **time-invariant**.

Solved Example 9: Testing Time-Invariance [GATE/ESE]

(a) $y(t) = x(t - 2)$ (pure delay)

Let $x_1(t) = x(t - t_0)$. Then $y_1(t) = x_1(t - 2) = x(t - t_0 - 2) = y(t - t_0)$. **Time-invariant.** ✓

(b) $y(t) = x(2t)$ (compressor)

$y_1(t) = x(2t - 2t_0)$ but $y(t - t_0) = x(2(t - t_0)) = x(2t - 2t_0)$. Wait — these are equal! But let us be careful: if input is $x(t - t_0)$, output is $x(2t - t_0)$. Compare with $y(t - t_0) = x(2t - 2t_0)$. Since $x(2t - t_0) \neq x(2t - 2t_0)$ in general, the system is **time-varying**. ✗

(c) $y(t) = e^{-t}x(t)$

Explicit t multiplying $x \Rightarrow$ **time-varying**. ✗

5.3 Causality

Causal System

A system is **causal** if $y(t_0)$ depends *only* on $x(t)$ for $t \leq t_0$ (present and past inputs; not future inputs):

$$y(t_0) = f(x(t), t \leq t_0)$$

A system using future inputs ($t > t_0$) is **non-causal**.

Causality in Practice

- All **physically realisable, real-time** systems must be causal.
- **Non-causal** systems can be implemented *offline* (e.g. image processing, data compression) where the entire signal is stored.
- In LTI systems: causal $\Leftrightarrow h(t) = 0$ for $t < 0$.

Solved Example 10: Causality [GATE/ESE]

- (a) $y(t) = x(t-1)$ (delay by 1). Output uses x at $t-1 \leq t$. **Causal.** ✓
 (b) $y(t) = x(t+1)$ (advance by 1). Output uses x at $t+1 > t$. **Non-causal.** ×
 (c) $y[n] = x[n] - x[n+2]$. Uses $x[n+2]$ (future). **Non-causal.** ×
 (d) $y[n] = \frac{1}{3}(x[n+1] + x[n] + x[n-1])$ (moving average). Uses $x[n+1]$ (future). **Non-causal.** × (Note: a causal version is $y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$.)

5.4 BIBO Stability**BIBO Stability**

A system is **BIBO (Bounded-Input Bounded-Output) stable** if every bounded input produces a bounded output:

$$|x(t)| \leq M_x < \infty \Rightarrow |y(t)| \leq M_y < \infty$$

for some finite constants M_x, M_y .

BIBO stability condition for LTI systems (derivation in Module 2):

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty \quad (\text{CT}) \quad \sum_{n=-\infty}^{\infty} |h[n]| < \infty \quad (\text{DT})$$

The impulse response $h(t)$ must be **absolutely integrable** (CT) or **absolutely summable** (DT).

Solved Example 11: BIBO Stability [GATE/ESE]

- (a) $h(t) = e^{-2t}u(t)$. $\int_0^{\infty} e^{-2t} dt = 1/2 < \infty$. **BIBO stable.** ✓
 (b) $h(t) = e^{2t}u(t)$. $\int_0^{\infty} e^{2t} dt = \infty$. **BIBO unstable.** ×
 (c) $h(t) = u(t)$ (ideal integrator). $\int_0^{\infty} 1 dt = \infty$. **BIBO unstable.** × (A constant input gives a ramp output $\rightarrow \infty$.)

5.5 Memory**Memoryless vs. System with Memory**

A system is **memoryless** if $y(t_0)$ depends *only* on $x(t_0)$ (the input at that same instant). If it depends on past or future input values, the system has **memory**.

Examples:

- Resistor: $y(t) = Rx(t)$ — **memoryless**.
- Capacitor: $y(t) = \frac{1}{C} \int_{-\infty}^t x(\tau) d\tau$ — **memory** (stores charge).

- Delay: $y[n] = x[n - 1]$ — **memory** (stores one previous sample).
- Accumulator: $y[n] = \sum_{k=-\infty}^n x[k]$ — **memory**.

Memory and Causality Relationship

- Every **memoryless** system is **causal** (output depends only on present input, which satisfies the causality condition).
- Not every causal system is memoryless (a delay is causal but has memory).

5.6 Invertibility

Invertible System

A system is **invertible** if distinct inputs always produce distinct outputs (one-to-one input-output mapping). Equivalently, there exists an *inverse system* $\mathcal{T}^{-1}\{\cdot\}$ such that:

$$\mathcal{T}^{-1}\{\mathcal{T}\{x(t)\}\} = x(t)$$

Examples:

- $y(t) = 2x(t)$: invertible; inverse is $x(t) = y(t)/2$.
- $y(t) = x^2(t)$: **not invertible**; $+x$ and $-x$ give identical output.
- $y[n] = x[n - 5]$ (delay by 5): invertible; inverse is $y[n + 5]$.
- Ideal integrator: $y(t) = \int_{-\infty}^t x(\tau)d\tau$; inverse is differentiation.

Invertibility in Communications

Channel equalisers and decoders exploit invertibility. The goal of channel coding is to design a system such that even a noisy channel's effect can be inverted (within limits) at the receiver.

5.7 Summary Table: System Properties

Property	Condition for YES	Fails if	Example
Linearity	Superposition holds; zero input $\Rightarrow$ zero output	Non-linear terms, $c \neq 0$ offset, non-zero IC	$\frac{dy}{dt} + ay = x(t)$
Time-Invariance	Constant coefficients; no explicit t	Explicit t , time-varying gain, $x(at)$ input	$y(t) = x(t-3)$
Causality	Output uses only past & present input	Output uses future x	$y(t) = x(t-1)$
BIBO Stability	$\int h dt < \infty$ or $\sum h < \infty$	Unbounded output for bounded input	$h(t) = e^{-t}u(t)$
Memoryless	Output depends only on current input	Depends on past or future	$y(t) = Rx(t)$
Invertibility	One-to-one mapping	Two inputs $\rightarrow$ same output	$y(t) = 2x(t)$

LTI Systems: The Most Important Class [GATE/ESE]

A system that is both **Linear** and **Time-Invariant** is called an **LTI system**. This is the central class studied in this course because:

- It is completely characterised by its **impulse response** $h(t)$ or $h[n]$.
- Output is given by **convolution**: $y(t) = x(t) * h(t)$.
- Analysis in the frequency domain (Fourier, Laplace, Z-transform) is straightforward.
- Most practical engineering systems are well-approximated as LTI.

Module 1 — One-Word Quick Recall

$\delta(t)$	Sifting	$u(t)$	Switch-ON
$e^{j\omega_0 t}$	Unit-Power / Eigenfunction	$A \cos(\omega_0 t)$	Power = $A^2/2$
Energy signal	Finite-Duration	Periodic	Power signal
LTI	Linear + TI	Causal	No-Future
Memoryless	Resistor-Like	Stable	Abs-Integrable $h(t)$
$x(at + b)$	Shift-then-Scale	Feedback	Stability/Control
Invertible	One-to-One	$\delta(at)$	$\frac{1}{ a }\delta(t)$

Module 1 — Key Formulae Reference Sheet

Signal Energy and Power

$$\text{CT Energy: } E_{\infty} = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad \text{CT Power: } P_{\infty} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$\text{DT Energy: } E_{\infty} = \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad \text{DT Power: } P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Elementary Signals

$$\text{Unit Step (CT): } u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \quad \text{Unit Ramp: } r(t) = t u(t)$$

$$\text{Impulse (CT): } \int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0) \quad \delta(at) = \frac{1}{|a|} \delta(t)$$

$$\text{Unit Sample (DT): } \delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \quad u[n] - u[n-1] = \delta[n]$$

$$\text{Product rule: } x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0) \quad \frac{d}{dt} u(t) = \delta(t)$$

Even–Odd Decomposition

$$x_e(t) = \frac{x(t) + x(-t)}{2} \quad (\text{Even part}) \quad x_o(t) = \frac{x(t) - x(-t)}{2} \quad (\text{Odd part})$$

Euler's Formula & Sinusoids

$$e^{j\theta} = \cos \theta + j \sin \theta \quad \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2} \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\text{Power of } A \cos(\omega_0 t + \phi): P = A^2/2 \quad \text{Power of } Ae^{j\omega_0 t}: P = |A|^2$$

$$\text{Power of } A + B \cos(\omega_0 t): P = A^2 + B^2/2$$

DT Periodicity Condition

$$x[n] = \cos(\omega_0 n) \text{ is periodic iff } \frac{\omega_0}{2\pi} = \frac{m}{N} \text{ is rational. Fundamental period: } N_0 = \frac{2\pi m}{\omega_0} \\ (\text{smallest positive integer } N_0).$$

End of Module 1